

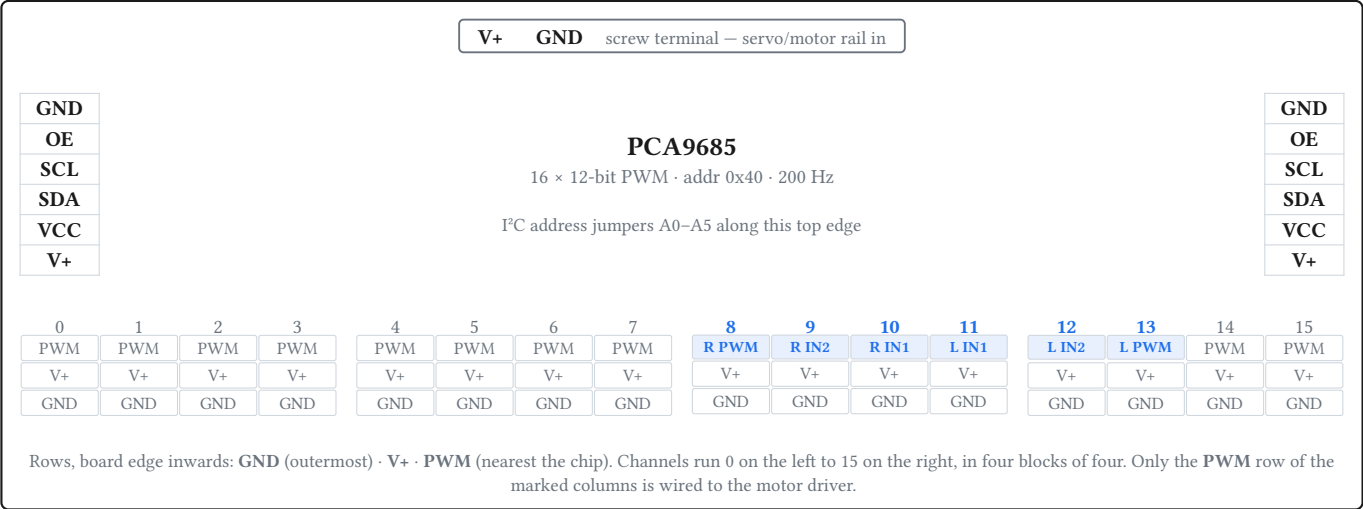
# Motor wiring card PCA9685 → TB6612FNG · robocar-unified

## Six signal wires, plus one that does not come from the PCA9685

Channels 8–13 carry direction and speed for both motors. **STBY** is the seventh wire and runs straight from the XIAO’s **D0 / GPIO1** — it is not a PCA9685 channel. With STBY low the H-bridges stay disabled and the motors do nothing, however correct the other six wires are.

## 1 · Board layouts, viewed from the component side

### PCA9685 16-channel PWM driver — Adafruit #815 and its clones



## Both side headers carry the same six pins

The 6-pin headers on the left and right edges are identical — GND, OE, SCL, SDA, VCC, V+ top to bottom — so either one can take the I<sup>2</sup>C feed and the other chains onward. Here the I<sup>2</sup>C side comes from **TCA9548A channel 0**, not from the XIAO directly. The board’s own underside warns: **the screw terminal is reverse-polarity protected, the side breakout pins are not** — the protection MOSFET sits behind the terminal only, so V+ fed backwards through a side header kills the board. **OE** needs nothing: a 10 kΩ pulldown on the board already enables the outputs.

### TB6612FNG dual motor driver — SparkFun ROB-14451 and its clones



## A01 / A02 / B01 / B02 are printed with a zero, not a letter O

Channel **A** drives the right motor and **B** the left, matching docs/schematics/circuits/robocar\_unified.py and the MOTOR\_RIGHT\_\* / MOTOR\_LEFT\_\* names in main/pin\_config.h. Nothing in the firmware can tell which motor is which — that is set entirely by which output pair you solder to which wheel.

## 2 · The seven wires

Rows are in the motor driver's own pin order, top to bottom, so working down this table is working down the header. Channel numbers ascend with it — that is the point of the assignment, not a coincidence.

|   | TB6612 pin      | From                          | Carries                                                                                          |
|---|-----------------|-------------------------------|--------------------------------------------------------------------------------------------------|
| □ | 1 · <b>PWMA</b> | PCA9685 ch 8 · PWM row        | Right motor speed — the only modulated line of the three                                         |
| □ | 2 · <b>AIN2</b> | PCA9685 ch 9 · PWM row        | Right motor direction bit 2                                                                      |
| □ | 3 · <b>AIN1</b> | PCA9685 ch 10 · PWM row       | Right motor direction bit 1 — driven full-on (4096) or full-off (0), never a duty cycle          |
| □ | 4 · <b>STBY</b> | XIAO <b>D0</b> / <b>GPIO1</b> | Global enable, straight from the MCU — the one wire in this column that is not a PCA9685 channel |
| □ | 5 · <b>BIN1</b> | PCA9685 ch 11 · PWM row       | Left motor direction bit 1                                                                       |
| □ | 6 · <b>BIN2</b> | PCA9685 ch 12 · PWM row       | Left motor direction bit 2                                                                       |
| □ | 7 · <b>PWMB</b> | PCA9685 ch 13 · PWM row       | Left motor speed                                                                                 |
| — | 8 · <b>GND</b>  | —                             | Redundant on-board with the other GND; leave it                                                  |

### The A side is PWM-then-direction, the B side direction-then-PWM

Not an inconsistency — the header is symmetric about STBY rather than repeated, so following it in order is what removes the crossings. If the channel numbers ever stop ascending down this table, the wiring and `main/pin_config.h` have diverged.

### Leave the V+ and GND rows of those six columns unconnected

Each PCA9685 channel breaks out three pins, but the motor driver takes only the signal. The V+ and GND pins of columns 8–13 stay empty; the driver gets its power and its ground from the rails below, not from the PWM header.

## 3 · Power and ground

|   | Pin                                | Rail                  | Note                                                                                                                     |
|---|------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| □ | TB6612 <b>VM</b>                   | 5.0 V from the LM2596 | Motor supply. Star-wire it from the regulator's output terminal — do not daisy-chain off the amplifier or the servo feed |
| □ | TB6612 <b>VCC</b>                  | 3.3 V                 | Logic supply. Must match the PCA9685's VCC — see §4                                                                      |
| □ | PCA9685 <b>VCC</b>                 | 3.3 V                 | Logic supply, via the side header                                                                                        |
| □ | PCA9685 <b>V+</b> (screw terminal) | 5.0 V from the LM2596 | Servo rail. 6 V absolute maximum, per the board's own silkscreen. Fit $\geq 470 \mu\text{F}$ here                        |
| □ | TB6612 <b>GND</b> (either pin)     | Common ground         | Both GND pins are joined on-board; one wire is enough                                                                    |
| □ | PCA9685 <b>GND</b>                 | Common ground         | Terminal-block GND and header GND are the same net                                                                       |

### A missing ground does not stop a chip from running

A board whose GND is not connected finds a return path through the ESD clamp diodes on its signal pins and keeps working — badly, in a way that reads as a firmware bug. Buzz every ground to the regulator's negative terminal **before** applying power, not after something misbehaves.

## 4 · Why both logic rails are 3.3 V

Both parts specify their input threshold as a fraction of **their own** supply, so the rail is not a free choice:

| Part            | Spec                         | Consequence                                                                                                                     |
|-----------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| PCA9685 SCL/SDA | $V_{IH} = 0.7 \times V_{DD}$ | At $V_{DD} = 5 \text{ V}$ the threshold is 3.5 V, above what the XIAO's 3.3 V I <sup>2</sup> C can drive. At 3.3 V it is 2.31 V |

| Part               | Spec                         | Consequence                                                                                        |
|--------------------|------------------------------|----------------------------------------------------------------------------------------------------|
| TB6612 IN1/IN2/PWM | $V_{IH} = 0.7 \times V_{CC}$ | At $V_{CC} = 5\text{ V}$ the threshold is 3.5 V, above the 3.3 V the PCA9685 would output          |
| TB6612 STBY        | $V_{IH} = 0.7 \times V_{CC}$ | At $V_{CC} = 5\text{ V}$ a 3.3 V GPIO cannot reliably lift STBY — the motors would stay in standby |

### WIRING.md currently says 5 V here — the schematic says 3.3 V

WIRING.md's power diagram feeds 5 V to both VCC pins; the schematic at docs/schematics/circuits/robocar\_unified.py feeds 3.3 V. The arithmetic above settles it in favour of 3.3 V for both, with the 5 V rail reaching only TB6612 VM and PCA9685 V+. Wire it that way and treat the diagram as the stale copy.

## 5 · Before power-up, and first motion

| Check                                                                                   | Expected                                                                                            |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Continuity, every GND pin to the regulator's negative terminal | Beeps — all of them                                                                                 |
| <input type="checkbox"/> Continuity, TB6612 VM to TB6612 VCC                            | <b>No</b> beep. A beep means the 5 V and 3.3 V rails are shorted                                    |
| <input type="checkbox"/> Regulator output, unloaded, before anything is connected       | 5.0 V — the LM2596 trimpot spans 1.2–37 V and arrives set at neither                                |
| <input type="checkbox"/> servo on the console                                           | Reports the live PWM frequency — proves the PCA9685 answers on TCA9548A ch 0 before any motor moves |
| <input type="checkbox"/> F on the console, wheels off the ground                        | Both wheels forward for about a second, then stop                                                   |
| <input type="checkbox"/> trace on the console                                           | I <sup>2</sup> C op counter climbing while driving, near-idle when stopped                          |

### Truth table, per channel

| IN1 | IN2 | PWM | Mode         |
|-----|-----|-----|--------------|
| H   | L   | H   | CW           |
| L   | H   | H   | CCW          |
| H   | L   | L   | Short brake  |
| H   | H   | —   | Short brake  |
| L   | L   | H   | Stop (coast) |

STBY must be high for any of it. The firmware drives forward as IN1 = high, IN2 = low.

### If a wheel runs backwards

Swap that motor's **two output wires** at A01/A02 or B01/B02. Do not swap the IN1/IN2 wires and do not change the firmware: the direction bits are shared with the reactive controller's turn logic, so inverting them there fixes driving straight and breaks turning.

### If nothing moves at all

Measure STBY. Below 2.31 V with  $V_{CC}$  at 3.3 V, the bridges are held off and every other wire is irrelevant.

**Sources.** Channel numbers and the STBY GPIO: main/pin\_config.h via docs/auto/pin\_defs.typ, regenerated and diffed by .github/workflows/build-guide-check.yml. Board layouts are read out of the vendors' published Eagle files — Adafruit Adafruit-16-Channel-PWM-Servo-Driver-PCB (Adafruit PCA9685 rev C.brd) and SparkFun Motor\_Driver-Dual\_TB6612FNG (SparkFun\_Motor\_Driver-TB6612FNG\_v11.brd) — not from a product photo. Thresholds: NXP PCA9685 datasheet rev. 4 and Toshiba TB6612FNG datasheet, both listed in docs/reference/datasheets/. Rails and decoupling: WIRING.md § Power.